```
> restart:with(DEtools):with(plots):with(linalg):
> #1
> #a
> ec_dif:=x*diff(y(x),x)-y(x)-ln(x^2+1)+Pi/2=0;
```

$$ec\_dif := x\left(\frac{d}{dx}\,y(x)\right) - y(x) - \ln\left(x^2+1\right) + \frac{1}{2}\,\pi = 0 \quad (1)$$

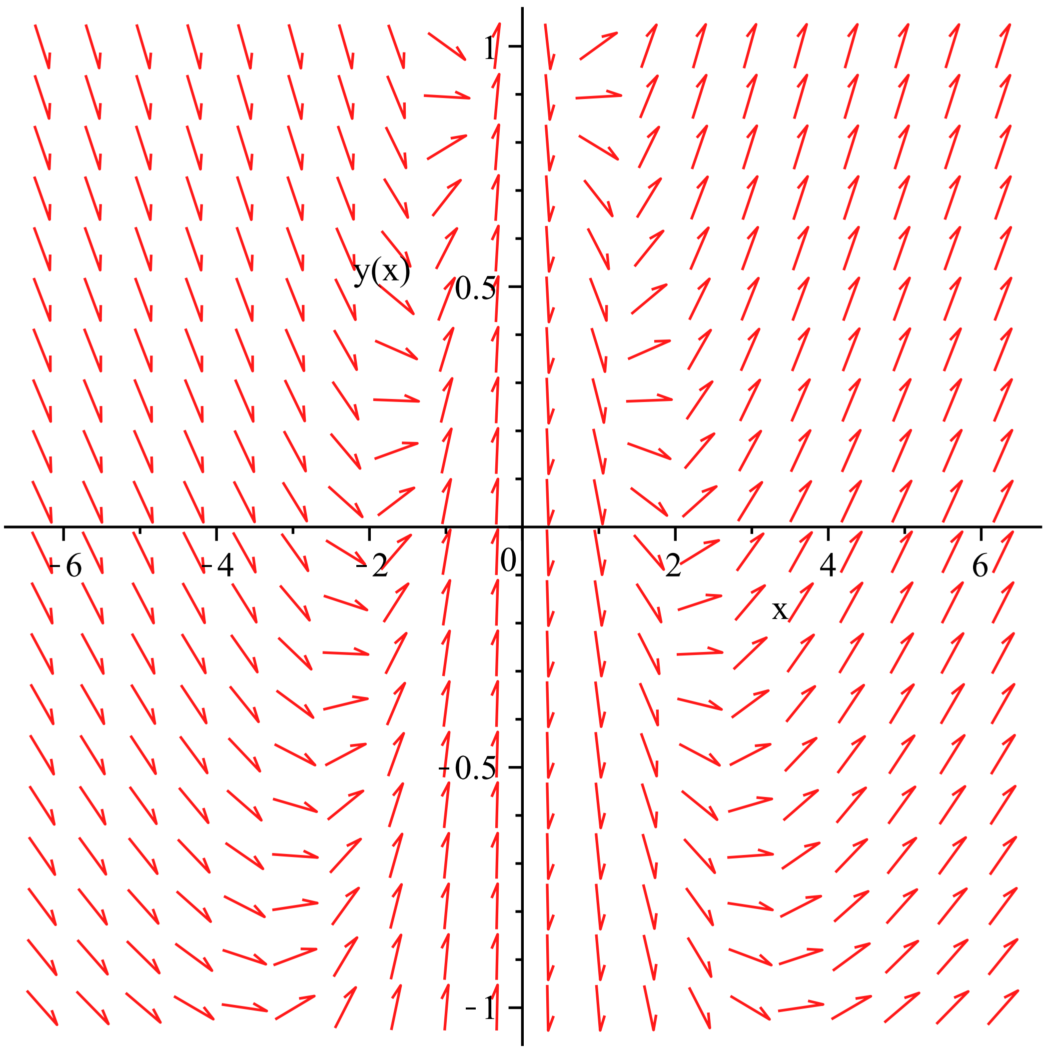
```
> sol:=dsolve(ec_dif,y(x));
```

$$sol := y(x) = -\ln\left(x^2+1\right) + 2\,x\arctan(x) + \frac{1}{2}\,\pi + x\,\_C1 \quad (2)$$

```
> grph:=unapply(rhs(sol),x);
```

$$grph := x \rightarrow -\ln\left(x^2+1\right) + 2\,x\arctan(x) + \frac{1}{2}\,\pi + x\,\_C1 \quad (3)$$

```
> DEplot(ec_dif,y(x),x=-2*Pi..2*Pi,y=-1..1);
```




```
> #b
> con_in:=y(1)=a;
```

$$con\_in := y(1) = a \quad (4)$$

```
> sol:=dsolve({ec_dif,con_in},{y(x)});
```

$$sol := y(x) = -\ln\left(x^2+1\right) + 2\,x\arctan(x) + \frac{1}{2}\,\pi + x\,(a + \ln(2) - \pi) \quad (5)$$

```
> grph:=unapply(rhs(sol),x,a);
```

$$grph := (x, a) \to -\ln(x^2+1) + 2\,x \arctan(x) + \frac{1}{2}\,\pi + x\,(a + \ln(2) - \pi) \tag{6}$$

```
> #2
> #a
> ec_dif_2:=diff(y(x),x,x)+3*diff(y(x),x)+2*y(x)=0;
```

$$ec\_dif\_2 := \frac{d^2}{dx^2}\,y(x) + 3\left(\frac{d}{dx}\,y(x)\right) + 2\,y(x) = 0 \tag{7}$$

```
> sol_2:=dsolve(ec_dif_2,y(x));
```

$$sol\_2 := y(x) = \_C1\,e^{-x} + \_C2\,e^{-2x} \tag{8}$$

```
> #b
> cond_in:=y(0)=3, D(y)(0)=2;
```

$$cond\_in := y(0) = 3, D(y)(0) = 2 \tag{9}$$

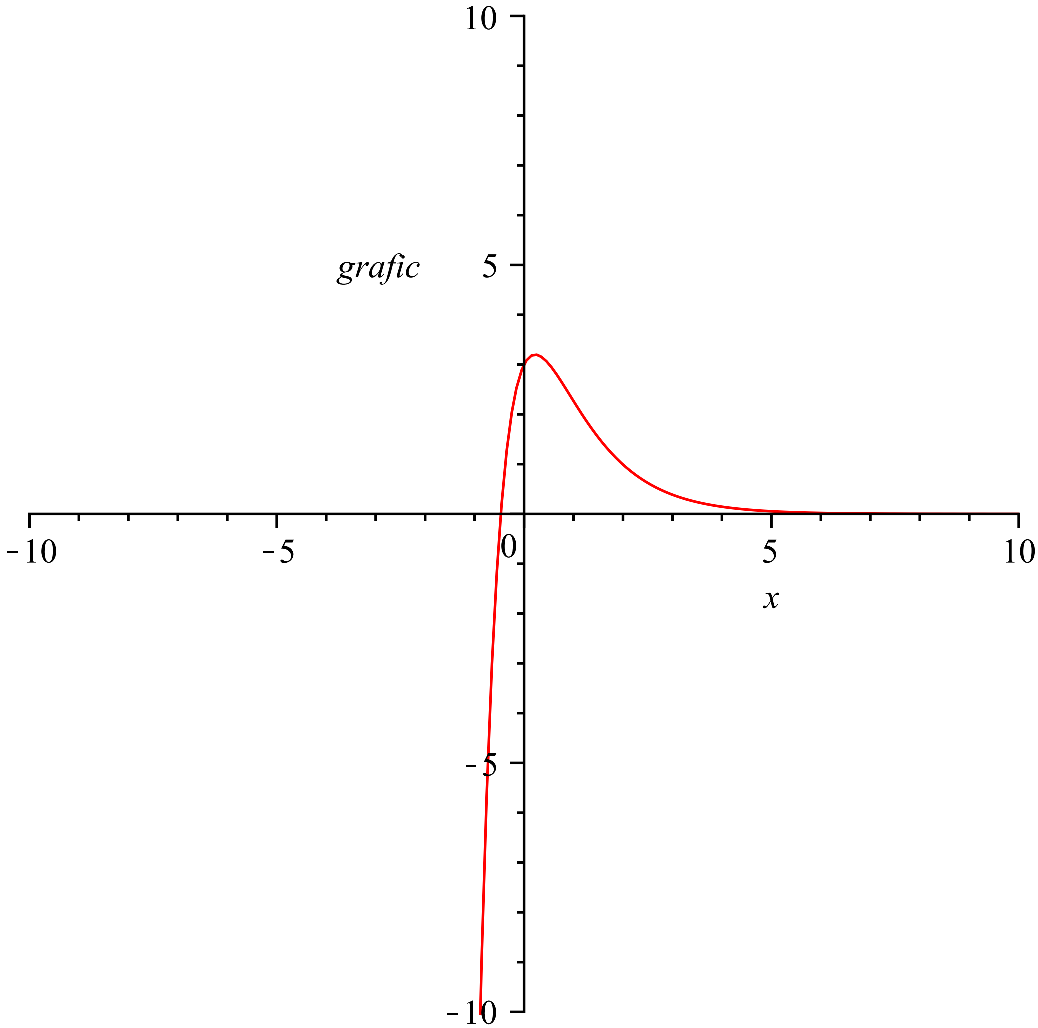
```
> sol_2_b:=dsolve({ec_dif_2,cond_in},{y(x)});
```

$$sol\_2\_b := y(x) = 8\,e^{-x} - 5\,e^{-2x} \tag{10}$$

```
> grafic:=unapply(rhs(sol_2_b),x);
```

$$grafic := x \to 8\,e^{-x} - 5\,e^{-2x} \tag{11}$$

```
> plot(grafic(x),x=-10..10,grafic=-10..10);
```




```
> #3
> #a
> ec1:=diff(x(t),t)=x(t)-y(t);
```

$$ec1 := \frac{d}{dt}\, x(t) = x(t) - y(t) \tag{12}$$

```
> ec2:=diff(y(t),t)=x(t)+y(t);
```

$$ec2 := \frac{d}{dt}\, y(t) = x(t) + y(t) \tag{13}$$

```
> sist:=ec1,ec2;
```

$$sist := \frac{d}{dt}\, x(t) = x(t) - y(t),\ \frac{d}{dt}\, y(t) = x(t) + y(t) \tag{14}$$

```
> sol_3:=dsolve({sist},{x(t),y(t)});
```

$$sol\_3 := \{x(t) = e^{t}\,(\_C1 \sin(t) + \_C2 \cos(t)),\ y(t) = e^{t}\,(-\_C1 \cos(t) + \_C2 \sin(t))\} \tag{15}$$

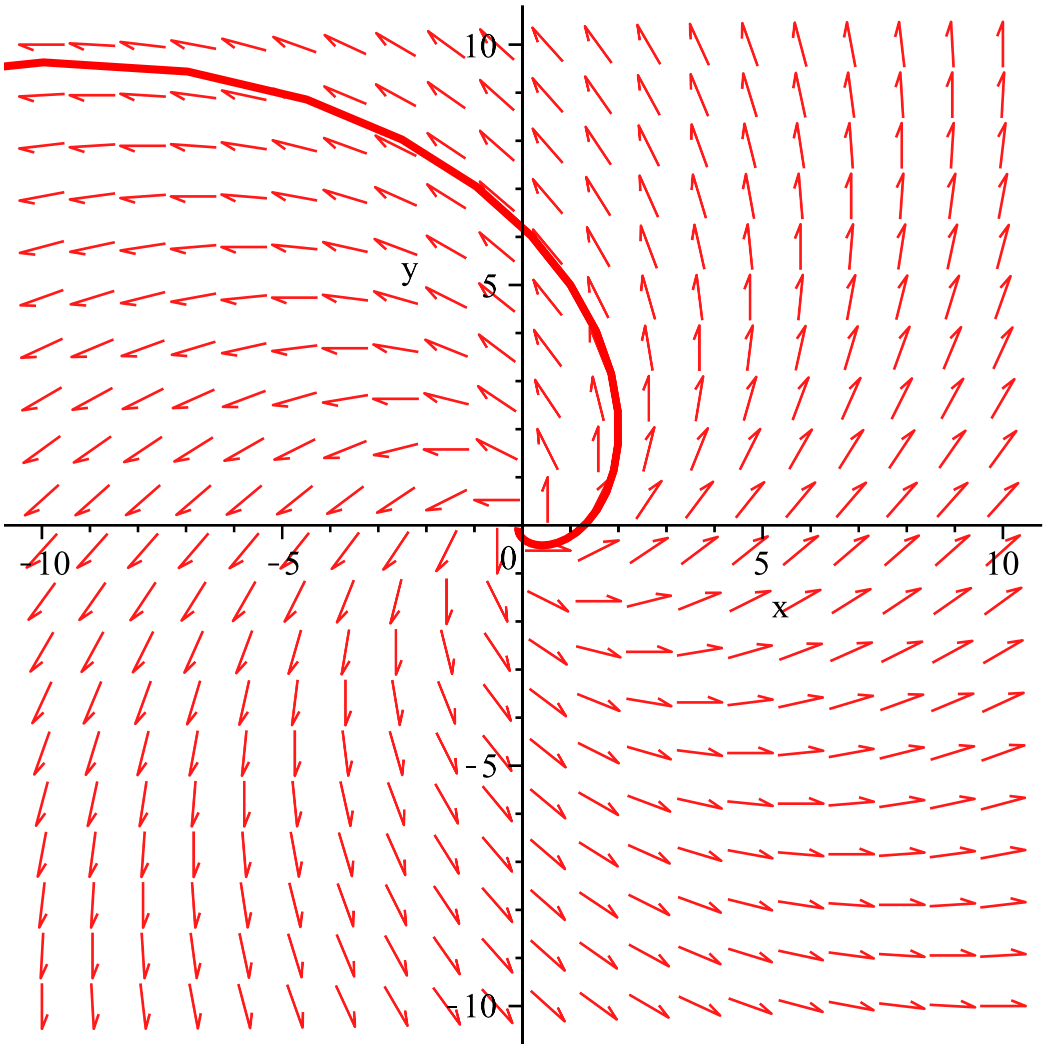
```
> A:=matrix([[1,-1],[1,1]]);
```

$$A := \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \tag{16}$$

```
> eigenvals(A);
```

$$1 + I,\ 1 - I \tag{17}$$

```
> #(0,0) este punct de echilibru instabil de tip focus
> #b
> DEplot([sist],[x(t),y(t)],t=-4..4,x=-10..10,y=-10..10,[[x(0)=0,y
  (0)=0],[x(0)=1,y(0)=5]],linecolor=[blue,red]);
```




```
> #limita(x(t),t->infinit)=limita(y(t),t->infinit)= 0?
> #nu deoarece (0,0) este punct de echilibru instabil, deci cand
  t->infinit, solutia se departeaza de 0
```

```
> #c
> sol_3:=dsolve({sist},{x(t),y(t)});
```

$$sol\_3 := \{x(t) = e^t (\_C1 \sin(t) + \_C2 \cos(t)), y(t) = e^t (-\_C1 \cos(t) + \_C2 \sin(t))\} \qquad (18)$$

```
> #d
> cond_in:=x(0)=-1,y(0)=1;
```

$$cond\_in := x(0) = -1, y(0) = 1 \qquad (19)$$

```
> sol_3_d:=dsolve({sist,cond_in},{x(t),y(t)});
```

$$sol\_3\_d := \{x(t) = e^t (-\sin(t) - \cos(t)), y(t) = e^t (\cos(t) - \sin(t))\} \qquad (20)$$

```
> #5
> f1:=(x,y)->x-y-1;
```

$$f1 := (x, y) \to x - y - 1 \quad (21)$$

```
> f2:=(x,y)->x^2*y+x*y;
```

$$f2 := (x, y) \to x^2 y + y x \quad (22)$$

```
> pct_echilibru:=dsolve({f1(x,y)=0,f2(x,y)=0},{x,y});
```

$$pct\_echilibru := [\{x=0\}, \{y=-1\}], [\{x=1\}, \{y=0\}], [\{x=-1\}, \{y=-2\}] \quad (23)$$

```
> J:=jacobian([f1(x,y),f2(x,y)],[x,y]);
```

$$J := \begin{bmatrix} 1 & -1 \\ 2\,y\,x + y & x^2 + x \end{bmatrix} \quad (24)$$

```
> A1:=subs(pct_echilibru[1,1],pct_echilibru[1,2],eval(J));
```

$$A1 := \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix} \quad (25)$$

```
> eigenvals(A1);
```

$$\frac{1}{2} + \frac{1}{2}\sqrt{5}, \frac{1}{2} - \frac{1}{2}\sqrt{5} \quad (26)$$

```
> evalf(1/2+1/2*sqrt(5));
```

$$1.618033988 \quad (27)$$

```
> evalf(1/2-1/2*sqrt(5));
```

$$-0.6180339880 \quad (28)$$

```
> #punctul de echilibru (0,-1) este punct instabil, tip sa
> A2:=subs(pct_echilibru[2,1],pct_echilibru[2,2],eval(J));
```

$$A2 := \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \quad (29)$$

```
> eigenvals(A2);
```

$$1, 2 \quad (30)$$

```
> #punctul de echilibru (1,0) este instabil de tip nod
> A3:=subs(pct_echilibru[3,1],pct_echilibru[3,2],eval(J));
```

$$A3 := \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \quad (31)$$

```
> valori_proprii:=eigenvals(A3);
```

$$valori\_proprii := \frac{1}{2} + \frac{1}{2}\,\mathrm{I}\sqrt{7}, \frac{1}{2} - \frac{1}{2}\,\mathrm{I}\sqrt{7} \quad (32)$$

```
> evalf(valori_proprii[1]);
```

$$0.5000000000 + 1.322875656\,\mathrm{I} \quad (33)$$

```
> evalf(valori_proprii[2]);
```

$$0.5000000000 - 1.322875656\,\mathrm{I} \quad (34)$$

```
> #punctul de echilibru (-1,-2) este punct de echilibru instabil,
  tip sa
> cond_in:=[x(0)=0,y(0)=i]$i=1..5,[x(0)=-i,y(0)=0]$i=1..5,[x(0)=0,y
  (0)=-i]$i=1..5,[x(0)=i,y(0)=0]$i=1..5;
```

$$cond\_in := [x(0)=0, y(0)=1], [x(0)=0, y(0)=2], [x(0)=0, y(0)=3], [x(0)=0, y(0) \quad (35)$$

$=4], [x(0)=0, y(0)=5], [x(0)=-1, y(0)=0], [x(0)=-2, y(0)=0], [x(0)=-3, y(0)=0], [x(0)=-4, y(0)=0], [x(0)=-5, y(0)=0], [x(0)=0, y(0)=-1], [x(0)=0, y(0)=-2], [x(0)=0, y(0)=-3], [x(0)=0, y(0)=-4], [x(0)=0, y(0)=-5], [x(0)=1, y(0)=0], [x(0)=2, y(0)=0], [x(0)=3, y(0)=0], [x(0)=4, y(0)=0], [x(0)=5, y(0)=0]$

```
> ec_dif_1:=diff(x(t),t)=f1(x(t),y(t));
```

$$ec\_dif\_1 := \frac{d}{dt}\, x(t) = x(t) - y(t) - 1 \tag{36}$$

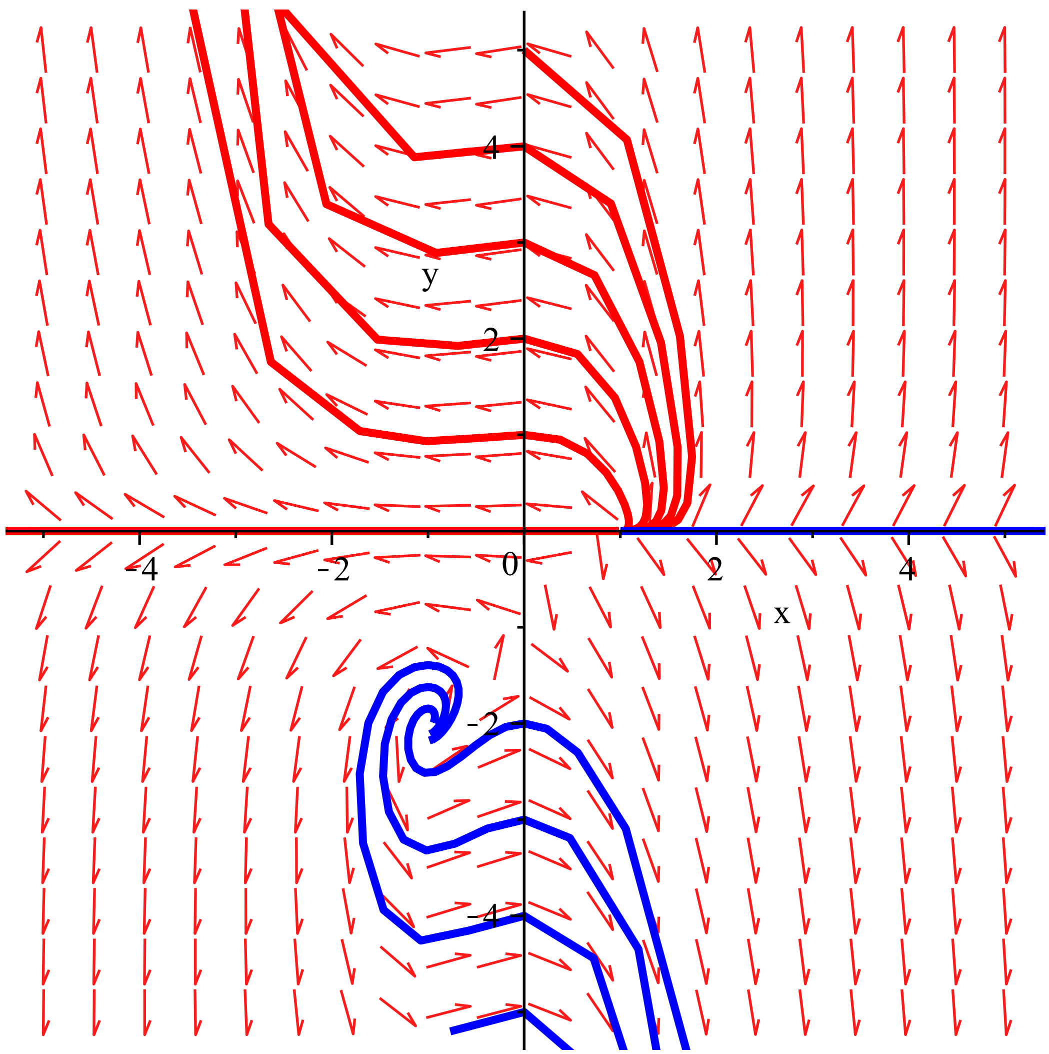
```
> ec_dif_2:=diff(y(t),t)=f2(x(t),y(t));
```

$$ec\_dif\_2 := \frac{d}{dt}\, y(t) = x(t)^2\, y(t) + y(t)\, x(t) \tag{37}$$

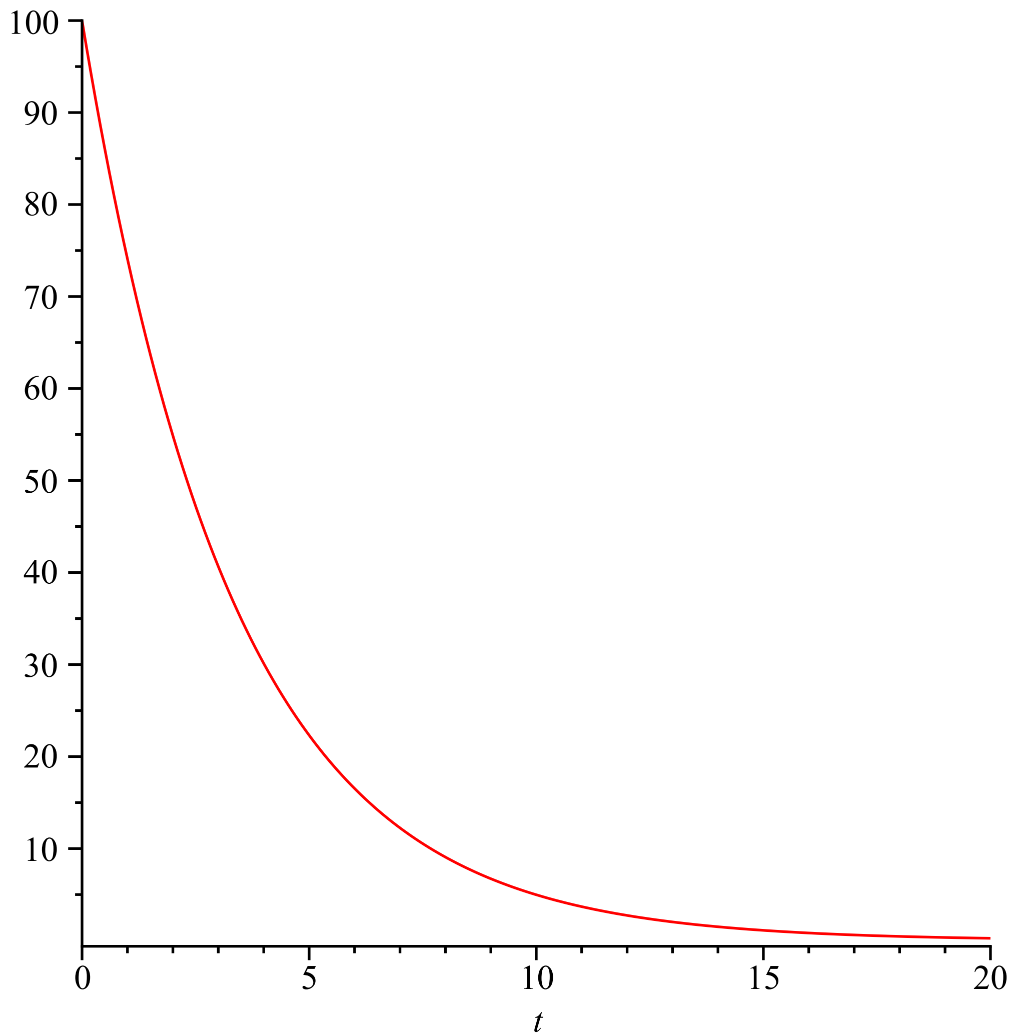
```
> sist:=ec_dif_1,ec_dif_2;
```

$$sist := \frac{d}{dt}\, x(t) = x(t) - y(t) - 1,\ \frac{d}{dt}\, y(t) = x(t)^2\, y(t) + y(t)\, x(t) \tag{38}$$

```
> DEplot([sist],[x(t),y(t)],t=-5..5,x=-5..5,y=-5..5,[cond_in],
  linecolor=[red$10,blue$10]);
```




```
> #4
```

```
> #a
> ec_1:=diff(x(t),t)=r*x(t);
```
$$ec\_1 := \frac{\mathrm{d}}{\mathrm{d}t}\, x(t) = r\, x(t) \quad (39)$$
```
> cond_in:=x(0)=x0;
```
$$cond\_in := x(0) = x0 \quad (40)$$
```
> solutie:=dsolve({ec_1,cond_in},x(t));
```
$$solutie := x(t) = x0\, \mathrm{e}^{r\,t} \quad (41)$$
```
> fct:=unapply(rhs(solutie),t,x0,r);
```
$$fct := (t, x0, r) \rightarrow x0\, \mathrm{e}^{r\,t} \quad (42)$$
```
> plot(fct(t,100,-0.30),t=0..20);
```



```
> #b
> ec_1:=fct(0,5*10^4,r)=5*10^4;
```
$$ec\_1 := 50000 = 50000 \quad (43)$$
```
> ec_2:=fct(7,5*10^4,r)=5.5*10^4;
```
$$ec\_2 := 50000\, \mathrm{e}^{7\,r} = 55000.0 \quad (44)$$
```
> rata:=solve({ec_1,ec_2},{r});
```
$$rata := \{r = 0.01361573997\} \quad (45)$$

```
> populatie_estimata:=fct(10,5*10^4,0.01361573997);
```

$$populatie\_estimata := 57293.11190 \quad (46)$$

```
> #populatia estimata dupa 10 ani va fi: 57293.11190
```